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New lithium-conducting nitride glass Li_3BN_2

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ABSTRACT

Novel nitride glass electrolyte Li_3BN_2 was prepared from Li_3N and BN *via* a mechanochemical process using planetary ball milling. Raman and X-ray photoelectron spectroscopies revealed that the glass was composed of Li^+ and BN_2^{3-} ions. The Li_3BN_2 glass exhibited good ductility, and the powder-compressed pellet showed a relative density of 84%. The hot-pressed pellet of the Li_3BN_2 glass showed a conductivity of $1.3 \times 10^{-5}\,S\,cm^{-1}$ at 25 °C, which is much higher than that of oxide-based glass electrolytes such as Li_3BO_3 glass and LiPON thin films. Moreover, Young's modulus of the Li_3BN_2 glass was 51.1 GPa, which is an intermediate value between sulfides and oxides. The Li symmetric cell using the Li_3BN_2 glass electrolyte was cycled stably at 100 °C without short-circuiting. Nitride glassy materials are promising electrolytes for all-solid-state batteries because of their high conductivity, good mechanical properties, and electrochemical stability.

1. Introduction

All-solid-state secondary batteries using nonflammable inorganic solid electrolytes have been expected to solve safety issues in commercially available lithium-ion batteries. For the fabrication of all-solid-state batteries, highly conducting solid electrolytes should be developed. Sulfide-based and oxide-based inorganic materials have been widely studied as promising electrolytes for all-solid-state batteries [1–8]. Some of the sulfide electrolytes were reported to show extremely high lithium ion conductivities of more than $10^{-2}\,S\,cm^{-1}$ at room temperature ($Li_7P_3S_{11}$ [1], $Li_{10}GeP_2S_{12}$ [2], $Li_{9.54}Si_{1.74}P_{1.44}S_{11.7}Cl_{0.3}$ [3], and $Li_{6+x}P_{1-x}Ge_xS_5I$ [4]). These conductivities are higher than those of typical organic liquid electrolytes. Moreover, sulfide electrolytes have good ductility, which enables their densification just by pressing at room temperature. Especially, glass electrolytes are more ductile than the corresponding crystalline ones, mainly due to the isotropic structure with free volume of the glasses [9]. This mechanical property is also the advantage to achieve favorable solid–solid contacts like electrode–electrolyte interfaces in all-solid-state batteries [10].

On the other hand, oxide-based materials have also attracted great attention because of their high chemical stability. Some of crystalline oxide electrolytes show high lithium ion conductivities of 10^{-4}–$10^{-3}\,S\,cm^{-1}$ at room temperature ($La_{0.51}Li_{0.34}TiO_{2.94}$ [5], $Li_7La_3Zr_2O_{12}$ [6], $Li_{1.3}Al_{0.3}Ti_{1.7}(PO_4)_3$ [7], and $LiTa_2PO_8$ [8]). However, these crystalline oxide electrolytes have poor ductility at room temperature, which generates an extremely high grain-boundary resistance. Thus, to achieve high densification leading to high lithium-ion conductivity, high temperature sintering at more than 1000 °C is needed in the case of $Li_7La_3Zr_2O_{12}$. In contrast to crystalline materials, glassy electrolytes tend to show better ductility. We prepared various oxide and sulfide glass electrolytes by melt-quenching and mechanochemical techniques [11–17]. Especially, by using a mechanochemical technique with a high-energy ball milling apparatus, glassy materials are prepared even in high alkaline content, and the conductivity of these glasses increases with the alkaline content [11,12]. For example, Li_3BO_3 ($75Li_2O25B_2O_3$ in mol%) glass with a high lithium content was prepared *via* mechanochemistry [11]. The prepared glass powder can be densified simply by pressing under 360 MPa at room temperature like ductile sulfide materials; the powder-compressed pellet showed a relatively high relative density of 71%, leading to a conductivity of $3.4 \times 10^{-7}\,S\,cm^{-1}$ at room temperature [11,13]. This good ductility for the glassy materials is favorable for constructing electrode–electrolyte interfaces in bulk-type all-solid-state batteries. However, the conductivity and ductility of oxide-based glassy electrolytes are not high enough, and thus various studies on the improvement of these properties of oxide glass materials were reported [14,15]. Nitrogen doping to Li_3PO_4 glass electrolytes is widely known to be effective in increasing their conductivity. By a partial substitute of nitrogen for oxygen in Li_3PO_4 glass thin films, the conductivity of glass electrolytes increased by one order of magnitude, and the lithium phosphorous oxynitride (LiPON) glass electrolyte exhibited a conductivity of about $10^{-6}\,S\,cm^{-1}$ at room temperature [18]. LiPON glass is practically used

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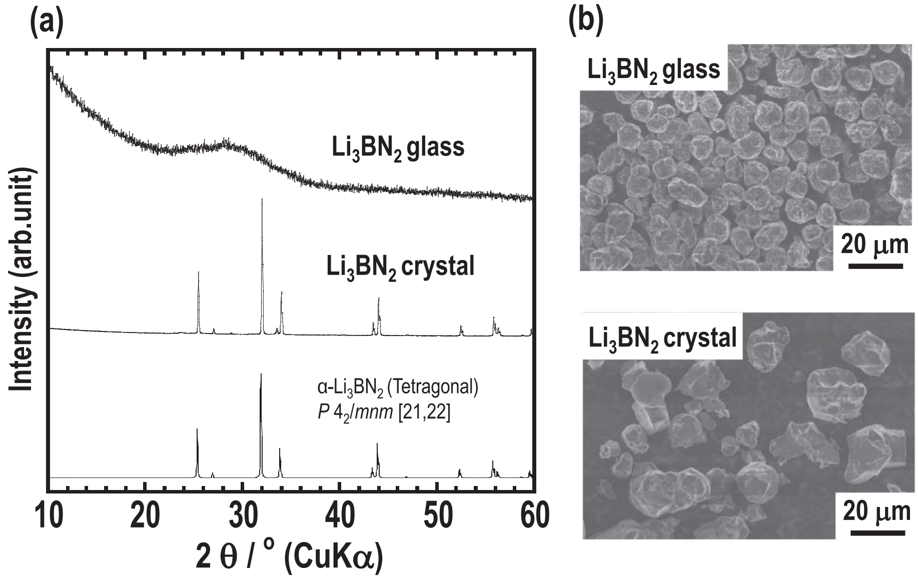

Fig. 1. (a) XRD patterns of the synthesized Li_3BN_2 glass and crystal. (b) SEM images of the Li_3BN_2 glass and crystal powders.

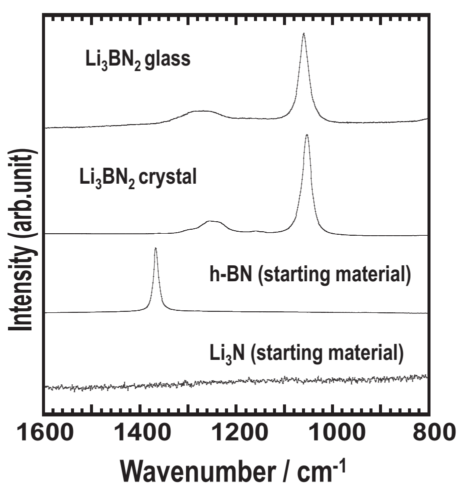

Fig. 2. Raman spectra of the Li_3BN_2 glass and crystal.

as an electrolyte in thin-film-type all-solid-state batteries. In oxynitride glasses, the glass transition temperature and the crystallization temperature increase with the nitrogen contents, which indicates that the thermal and chemical stabilities are improved by the substitution of nitrogen [19]; nitrogen is replaced with bridging oxygen to increase cross-linking of the glass network. Recently, we have reported that nitrogen doping is also effective in improving conductivity and chemical stability even of sulfide-based electrolytes [16]. Therefore, the incorporation of nitrogen to oxide or sulfide lithium-ion conductors is effective in enhancing performance of solid electrolytes.

However, there are few reports on nitride electrolytes. Crystalline nitride materials as lithium ion conductors were reported in systems Li–B–N, Li–Si–N, and Li–P–N [20], but they showed quite low conductivity; crystalline Li_3BN_2 electrolytes exhibited a conductivity of 3×10^{-7} S cm^{-1} even at 400 K [21]. On the other hand, lithium-ion conducting amorphous nitride materials has never been reported so far. Therefore, it is important to synthesize novel amorphous electrolytes and evaluate their properties. Here, we report the mechanochemical synthesis of Li_3BN_2 glass electrolyte. Glassy Li_3BN_2 samples were prepared by using planetary ball milling, and their local structure and thermal properties were investigated. In addition, the mechanical and electrical properties were also examined, and these properties were compared with those of Li_3BO_3 and Li_3PS_4 in terms of electronegativity of X in Li_3MX_a.

2. Experimental

2.1. Synthesis

A glassy Li_3BN_2 electrolyte was synthesized using mechanochemistry of Li_3N and BN crystals as follows. All processes were conducted in dry Ar atmosphere to avoid the influence of moisture and oxygen in the air. First, Li_3N was prepared from a metallic Li foil (99.9%, Furuuchi Chemical Corp.) and N_2 gas (6 N). The Li foil was placed into a stainless steel container, and then Ar gas (6 N) inside the container was replaced with N_2 gas. The container was sealed and left for 3 days at room temperature, and then Li_3N was formed.

Li_3BN_2 glass was mechanochemically synthesized by using a planetary ball milling apparatus (Pulverisette 7; Fritsch GmbH). A mixture of Li_3N and h-BN (98%, Aldrich Chemical Co. Inc.) was placed into a zirconia pot (internal volume of 45 ml) with zirconia balls (5 mm in diameter). The weight ratio of the sample to the ball was 1/65. The rotation speed and the milling period of time were respectively 310 rpm and 40 h. Contamination of zirconia in the milled samples was negligible as elemental zirconium was not detected by XPS.

A crystalline Li_3BN_2 electrolyte was prepared by a conventional solid-state reaction of Li_3N and BN in the way described in the previous report [20]. A mixture of the initial materials was compressed to the pellet form and then enclosed in a tantalum foil. After sintering at 800 °C for 1 h, the pellet was ground by agate mortar and pestle to obtain a crystal powder.

2.2. Characterization

X-ray diffraction (XRD) measurements were conducted with CuKα radiation using a diffractometer (SmartLab; Rigaku Corp.) to identify the structure of glass and crystal samples. Raman spectra were obtained using the 325 nm line of a He—Cd laser with a Raman spectrometer (LabRAM HR800; Horiba Ltd.). X-ray photoelectron spectroscopy (XPS) measurements were carried out for the Li_3BN_2 glass and crystal powders by using a spectrometer (K-Alpha, Thermo Fisher Scientific). The samples were transferred to the analysis chamber using a vessel packed with dry Ar gas to prevent the samples from air exposure. Monochromatic Al-Kα radiation (1486.6 eV) was used as the X-ray

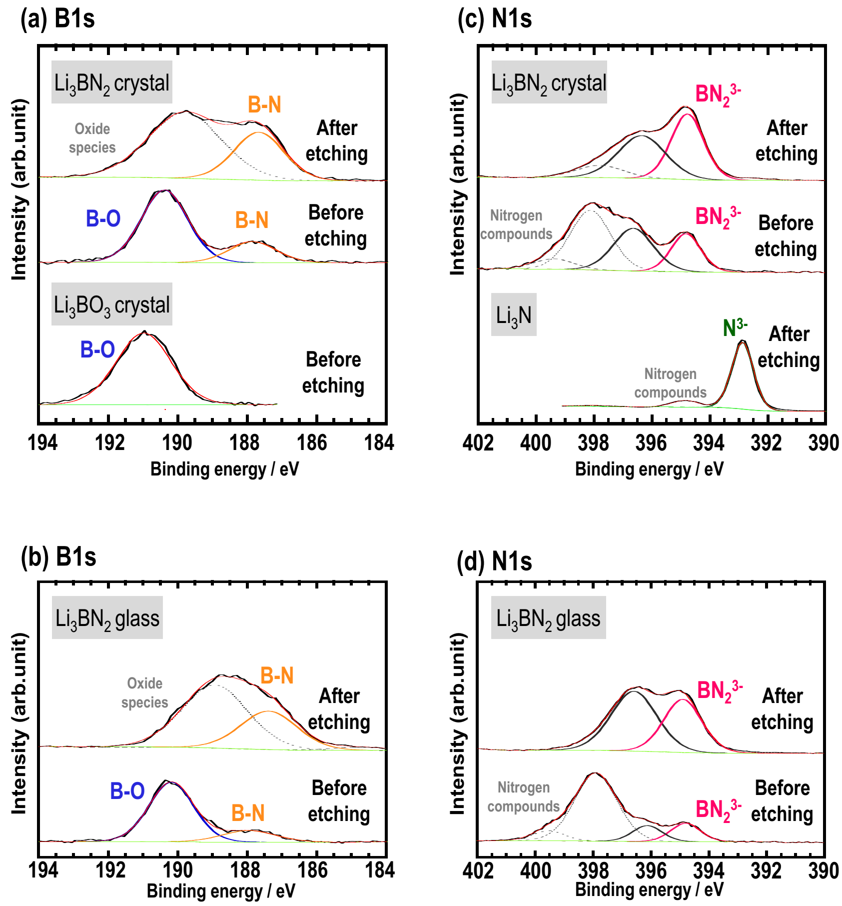

Fig. 3. XPS spectra of (a) B1s of the Li_3BN_2 crystal and Li_3BO_3 crystal, (b) B1s of the Li_3BN_2 glass, (c) N1 s of the Li_3BN_2 crystal and Li_3N, and (d) N1 s of the Li_3BN_2 glass.

source. To remove the influence of the charging effect on the spectra, the samples were neutralized using a flood gun during the measurements. The samples were etched by Ar^+ ion monomer for 3 min. The obtained spectra were calibrated by setting the measured binding energy of the C1s peak to 284.7 eV of adventitious carbon accumulated in the analysis chamber of the spectrometer.

Differential thermal analysis (DTA) was performed using a thermal analyzer (Thermo Plus EVO2; Rigaku Corp.) to determine the crystallization temperatures. The glass sample enclosed in an Al pan was heated at 10 °C for 1 min under the N_2 gas flow up to 550 °C.

The ionic conductivities were measured for the Li_3BN_2 glass and crystal. These powders were uniaxially pressed under 720 MPa at room temperature for 5 min. In addition, to obtain a further densified pellet, the hot-pressing technique at around the glass-transition temperature was carried out. The Li_3BN_2 glass powder was pressed under 360 MPa at 300 °C for 12 h. The pellet diameter and thickness were respectively 10 mm and approximately 1 mm. A gold thin-film electrode was deposited on both surfaces of the pellets by using Quick Coater (SC-701MkII; Sanyu Electron Corp.) placed in an Ar-filled glove box. Then, the conductivity was measured by the AC impedance method using an impedance analyzer (SI-1260; Solartron Analytical). The frequency range was from 0.1 Hz to 8 MHz, and the applied voltage was 50 mV. All the measurements were performed in Ar atmosphere.

The bulk density of the powder-compressed pellets was calculated from the weight and volume of the pellets. The apparent density of the Li_3BN_2 glass and crystal powders was measured by using an Ar gas pycnometer (AccuPycII1340; Shimadzu Corp.) placed in an Ar-filled glove box. In general, closed pores in particles may contribute to decreasing the apparent density. However, the apparent density (1.76 g cm^{-3}) of the synthesized Li_3BN_2 crystal was almost the same as the XRD density (1.75 g cm^{-3}), indicating that the influence of closed pores on the apparent density is almost neglected. Thus, the apparent density was treated as a theoretical one in this study. The apparent density of the glass containing free volume was smaller than that of the corresponding crystal. The relative density of the pellets was calculated by dividing the bulk density of the powder-compressed pellets by the apparent density of the powders. The morphology of the Li_3BN_2 glass and crystal powders was observed by a field emission scanning electron microscope (FE-SEM, SU8220; Hitachi High-Technologies Corp). Moreover, the cross-section of the powder-compressed pellet was also observed to investigate the ductility of the electrolyte. To obtain a smooth surface for the observation, the fracture of the pellet was polished using an Ar^+ ion milling system (IM4000; Hitachi High-Technologies Corp).

The mechanical property of the Li_3BN_2 glass was evaluated by an ultrasonic pulse-echo technique using 5 MHz frequency transducers. To avoid the influence of any voids in the pellet, the hot-pressed pellet was used for the measurement. The diameter and thickness of the prepared pellets were approximately 10 mm and 2 mm, respectively. The prepared pellets were enclosed in a polymer bag in an Ar-filled glove box and evaluated using an ultrasonic pulser/receiver (5077 PR, Olympus) in the air atmosphere. A zippered polyethylene plastic bag was used in order to avoid the exposure to regular air atmosphere. The influence of the bag was found to be negligible because the bag was sufficiently thin (0.04 mm) [10].

To investigate the electrochemical stability of the Li_3BN_2 glass electrolyte to the lithium metal, galvanostatic cycling tests using a Li symmetrical cell (Li/Li_3BN_2 glass/Li) were carried out. The Li_3BN_2 glass

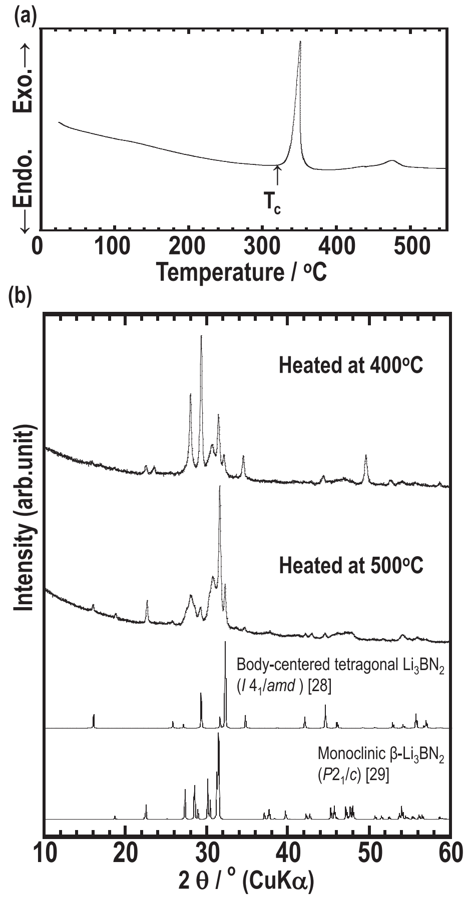

Fig. 4. (a) DTA curve of the Li_3BN_2 glass. (b) XRD patterns of the Li_3BN_2 glass-ceramics heated at 400 °C and 500 °C.

powder was uniaxially pressed into a pellet at 720 MPa. Lithium foils and cupper foils were attached on both sides of the pellet, and then pressed again at 80 MPa by using a cold isostatic pressing apparatus to form good contact between the electrolyte and the lithium metal. The electrochemical measurement was performed at 100 °C using a charge–discharge measurement device (BTS-2004, Nagano Co. Ltd.). The used current density and the time were 0.13 mA cm^{-2} and 1 h, respectively.

3. Results and discussion

The XRD patterns of the Li_3BN_2 glass and crystal are shown in Fig. 1a. A halo pattern was observed in the Li_3BN_2 glass prepared by the mechanochemical process. However, diffraction peaks attributable to α-Li_3BN_2 with the space group of $P4_2/mnm$ (136) [21,22] were observed for the Li_3BN_2 crystal prepared by the solid-state synthesis. In the following discussion, crystalline Li_3BN_2 (α-Li_3BN_2) is referred to as "crystal".

The Raman spectra of the Li_3BN_2 glass and crystal are presented in Fig. 2. The main peak was observed at around 1050 cm^{-1}, which is attributable to the Raman band due to the symmetric stretching in the BN_2^{3-} $(N{=}B{=}N)^{3-}$ ion unit [23]. The glass structure was mainly composed of isolated BN_2^{3-} anions and Li^+ cations, similar to the Li_3BN_2 crystal.

The SEM images of the Li_3BN_2 glass and crystal powders are shown in Fig. 1b. Particles with a uniform size of about 10 μm were observed in the glass. However, the particle size of the Li_3BN_2 crystal was not uniform in size (about 5–40 μm).

The bonding characters of the Li_3BN_2 glass and crystal were investigated by XPS. Fig. 3 presents the XPS spectra for B1s and N1s of the Li_3BN_2 glass and crystal before and after etching. In both B1s and N1s spectra, there was a slight difference between the peak positions of the glass and crystal. The B1s spectra of the Li_3BN_2 glass and crystal in Fig. 3a and b show two peaks around 190.5 eV and 187.5 eV. The peak at 190.5 eV is attributable to the B—O bond because Li_3BO_3 composed BO_3^{3-} anions gave the peak at 191.0 eV in the B1s spectrum (Fig. 3a).

Since the peak assigned to the BN_2^{3-} ion was observed in the Raman spectra (Fig. 2), another peak at 187.5 eV can be attributed to the B—N bond. The peak intensity derived from the B—N bond increased relatively to that derived from the oxides after Ar^+ etching, indicating that the main structure of the particle is nitrides.

In the N1s spectra of the Li_3BN_2 glass and crystal shown in Fig. 3c and d, two main peaks were observed at 394.7 eV and 396.2 eV. Considering that the peak attributable to B—N bond was observed in the B1s spectra shown in Fig. 3a, the peak assigned to the nitrogen anions bonding to boron should be observed. In the N1 s spectra of Li_3N, the peak attributable to N^{3-} was observed at 393 eV (Fig. 3c). In addition, the binding energies for the bridging nitrogen of N_d (–N =) and N_t (–N <) are reported to be at 397.0 eV and 398.6 eV, respectively [24,25]. On the other hand, Lacivita et al. mentioned the presence of apical nitrogen (non-bridging N) and proposed that N 1 s peaks at 397.8 eV and 399.2 eV for LiPON are attributable to N_a (apical N) and N_d, respectively [26]. Anyway, the two peaks at 394.7 eV and 396.2 eV for the Li_3BN_2 samples were located at the intermediate binding energy region between the N^{3-} anion and the bridging nitrogen. It is thus suggested that the peak at 394.7 eV is attributable to the non-bridging nitrogen in the BN_2^{3-} anion, and the other peak is possibly assigned to the nitrogen in the BN_2^{3-} anion partially affected by oxygen as an impurity. Further structural analyses are needed to be done for clarifying the origin of the doublet peaks in N1s spectra observed in this study.

Fig. 4a shows the DTA curve of the Li_3BN_2 glass. Two exothermic peaks were observed at around 310 °C and 450 °C, which are higher than the crystallization temperature of Li_3BO_3 glass (240 °C) [11]. The crystallization behavior of the Li_3BN_2 glass was investigated by heating the glass at the temperatures beyond these exothermic peaks. In the following discussion, crystallized glass samples are referred to as "glass-ceramics". Fig. 4b describes the XRD patterns of the Li_3BN_2 glass-ceramics heat-treated at 400 °C and 500 °C. The pattern of the Li_3BN_2 glass-ceramics heated at 400 °C was not assigned to any reported crystalline structures. A new thermodynamically metastable crystalline phase (unknown phase) is obtained. Metastable phases are often discovered when glass is heated above the crystallization temperature [14,27]. By heating the glass at 500 °C, the pattern attributable to body-centered tetragonal Li_3BN_2 [28] and new unknown peaks, which would be indexed to monoclinic Li_3BN_2 (β-Li_3BN_2) [29], appeared. The exothermic peak at around 450 °C can be assigned to the phase transition from the unknown phase to these two crystalline phases.

To investigate the ductility of the Li_3BN_2 electrolyte, green compacts of the glass and crystal powders were prepared. Fig. 5 shows the cross-sectional SEM images of the Li_3BN_2 glass (cold-pressed and hot-pressed) and the crystal (cold-pressed) pellets. The Li_3BN_2 glass was densified simply by pressing at 720 MPa at room temperature (Fig. 5a). The relative density of the powder-compressed pellet was 84%, which is higher than that of the Li_3BO_3 glass pressed at 720 MPa (79%). In the SEM image of the Li_3BN_2 crystal pellet (Fig. 5b), cracks and grain boundaries were clearly observed. Room temperature pressure sintering phenomena [10] occurred in the particle boundaries, and obvious grain boundaries observed in the crystalline Li_3BN_2 pellet were not noted for the glassy Li_3BN_2 pellets. The difference in the microstructure (amorphous or crystalline matrix) strongly affects the relative density for the compacts. Thus, the Li_3BN_2 glass has better formability (ductility) than the corresponding crystal. Better ductility of a glass electrolyte is favorable for preparing a dense electrolyte pellet and constructing favorable electrode–electrolyte interfaces in all-solid-state batteries. Glass becomes a supercooled liquid and shows a viscos flow under high

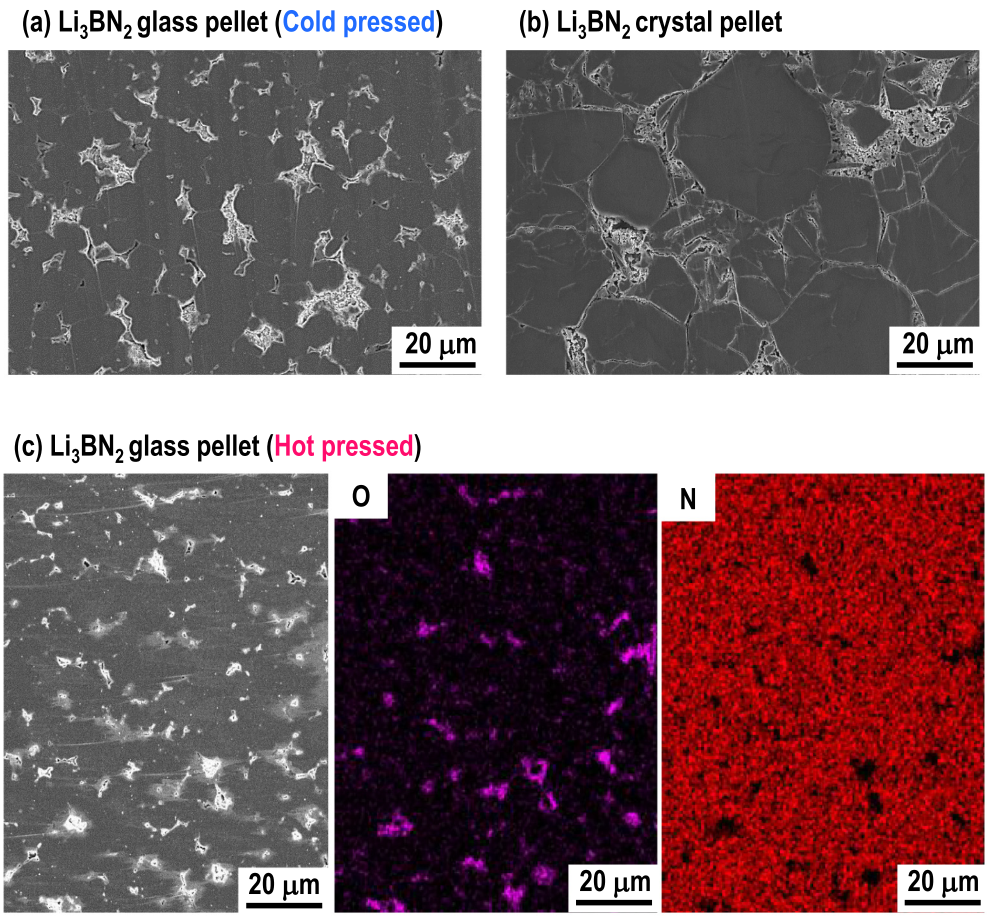

Fig. 5. Cross-sectional SEM images of (a) the cold-pressed Li_3BN_2 glass and (b) cold-pressed Li_3BN_2 crystal and (c) cross-sectional SEM image and EDX mapping of the hot-pressed Li_3BN_2 glass.

Table 1
Molding temperature, molding pressure, bulk density of the powder-compressed pellet (ρ), apparent density of powder (ρ_O), relative density (ρ/ρ_O), longitudinal velocity (V_L), shear velocity (V_S), shear modulus (G), Young's modulus (E), bulk modulus (K), and Poisson's ratio (ν) of Li_3BN_2 and Li_3BO_3. These of Li_3PS_4 are also shown for comparison.

	Molding temp./°C	Molding pressure/MPa	ρ/g cm^{-3}	ρ_O/g cm^{-3}	ρ/ρ_O/%	V_L/m s^{-1}	V_S/m s^{-1}	G/GPa	E/GPa	K/GPa	ν
Li_3BN_2	300	360	1.59	1.68	95	6370	3540	20.0	51.1	38.1	0.28
Li_3BO_3	230	540	1.90	2.07	92	7080	3560	24.1	64.2	63.2	0.33
Li_3PS_4 [9,10]	190	360	1.88	1.88	~100	4150	2150	8.7	22.9	20.8	0.32

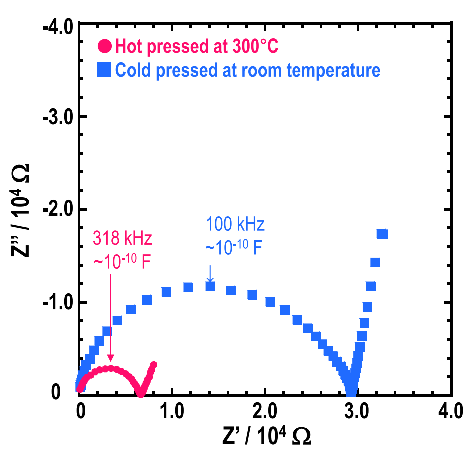

Fig. 6. Nyquist plots of the cold-pressed and hot-pressed Li_3BN_2 glass electrolytes at room temperature.

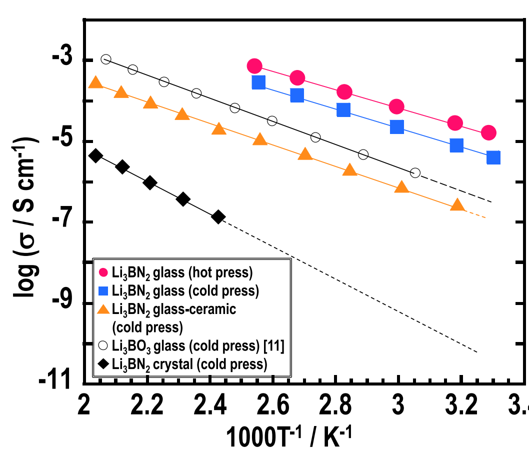

Fig. 7. Temperature dependences of the ionic conductivities of the Li_3BN_2 glass, glass-ceramics, and crystal. That of the Li_3BO_3 glass is also shown for comparison.

Table 2
Conductivities, activation energies, and relative densities of the Li_3BN_2 glass, glass-ceramics, and crystal.

	σ_{25} (S cm^{-1})	σ_{100} (S cm^{-1})	E_a (kJ mol^{-1})	Relative density (%)
Li_3BN_2 glass (cold press)	3.1×10^{-6}	1.4×10^{-4}	47	84
Li_3BN_2 glass (hot press)	1.3×10^{-5}	3.7×10^{-4}	41	95
Li_3BN_2 glass-ceramic	1.1×10^{-7}	5.6×10^{-6}	48	84
Li_3BN_2 crystal (this study)	2.9×10^{-11}	1.3×10^{-8}	76	83
Li_3BO_3 glass [11]	3.4×10^{-7}	1×10^{-5}	52	79

pressure above the glass transition temperature. Thus, highly dense pellets can be obtained by hot pressing [30]. As shown in Fig. 5c, a further densified Li_3BN_2 glass pellet was successfully obtained by hot pressing at 300 °C. The relative density was 95%. The EDX mappings of the hot-pressed Li_3BN_2 pellet are also shown in Fig. 5c. In the EDX mappings, a signal of nitrogen was entirely detected, while that of oxygen as an impurity was observed especially in voids.

Furthermore, the elastic property of the powder-compressed pellet of the Li_3BN_2 glass was evaluated. The elastic property is important for solid electrolytes to keep solid-state interfaces even when active materials undergo volume changes during charge–discharge processes [10,31]. The elastic moduli of the Li_3BN_2 glass are summarized in Table 1. Those of the Li_3BO_3 glass and Li_3PS_4 glass are also listed for comparison. These values were evaluated using the hot-pressed glass electrolytes. Young's modulus (E) of the Li_3BN_2 glass was 51.1 GPa, which was an intermediate value between Li_3PS_4 sulfide and Li_3BO_3 oxide. For lithium silicate oxynitride glasses, the elastic modulus was increased with the nitrogen content in oxide glasses because of increasing the glass network connectivity by replacing bridging oxygen with nitrogen [19]. However, the Li_3BN_2 glass showed lower elastic modulus than the corresponding Li_3BO_3 oxide glass because nitrogen exists in non-bridging states based on the results from the Raman and XPS analyses.

Conductivity measurements were carried out using the AC impedance method. The Nyquist plots for the cold-pressed and hot-pressed Li_3BN_2 glasses at 30 °C are shown in Fig. 6. Only one semicircle was observed in both of the Nyquist plots. The capacitance for this semicircle was calculated to be about 10^{-10} F. The semicircle includes both bulk and grain boundary components, and their separation was difficult. Thus, the conductivity of the electrolytes was calculated from the total resistance calculated from the intersection at the X-axis of the semicircle in the lower frequency region.

Fig. 7 presents the temperature dependence of the conductivities of the Li_3BN_2 glass electrolytes prepared by cold and hot pressing. Those of the Li_3BN_2 glass-ceramics heated at 400 °C, Li_3BN_2 crystal, and Li_3BO_3 glass are also shown for comparison. The temperature dependence of the conductivities obeyed the Arrhenius equation, and the activation energy for conduction was calculated. The results of the conductivity (at 25 °C and 100 °C) and activation energy are summarized in Table 2. The cold-pressed Li_3BN_2 glass showed a conductivity of 3.1×10^{-6} S cm^{-1} (25 °C) and 1.4×10^{-4} S cm^{-1} (100 °C) and an activation energy of 47 kJ mol^{-1}. This conductivity is about one order of magnitude higher and the activation energy is lower than those of the Li_3BO_3 glass electrolyte [11]. Moreover, by further densification by hot pressing, the conductivity increased to 1.3×10^{-5} S cm^{-1} at 25 °C and the activation energy decreased to 41 kJ mol^{-1}. It is considered that the conductivity increased and the activation energy decreased because the contribution of the grain boundary resistance decreased. As mentioned in the results of XPS (Fig. 3), the oxide species exist on the surface of the particle as an impurity phase, which would prevent the densification and ionic conduction. Thus, it is expected that removing surface oxygen will further improve the ductility and the conductivity of the cold-pressed pellet.

The Li_3BN_2 crystal (α-Li_3BN_2) exhibited a low conductivity of 2.9×10^{-11} S cm^{-1} at 25 °C. This is consistent with the previous report [21]. Therefore, the conductivity of Li_3BN_2 greatly increased after amorphization, and the Li_3BN_2 glass showed five orders of magnitude higher conductivity than the corresponding crystal. Moreover, we examined the crystallization of the Li_3BN_2 glass with the expectation of further improvement of the conductivity by the crystallization of highly conducting metastable phase [1,14,16,32,33]. Unfortunately, the conductivity of the Li_3BN_2 glass-ceramics heated at 400 °C was 1.1×10^{-7} S cm^{-1} at 25 °C, which was lower than that of the Li_3BN_2 glass. However, it is noted that the Li_3BN_2 glass-ceramics showed higher conductivity than the Li_3BN_2 crystal. Thus, the conductivity of the newly developed phase (unknown phase) should be higher than that of α-Li_3BN_2. The detailed structural analysis was not carried out for the glass-ceramic electrolytes. To understand the crystallization phenomena and the conduction property of the Li_3BN_2 glass-ceramic electrolytes, further structural analyses should be performed.

The Li_3BN_2 glass showed much higher conductivity than the LiPON thin film (2×10^{-6} S cm^{-1} at 25 °C) [18]. When considering the application to a thin-film-type all-solid-state battery, the conductivity which is one order of magnitude higher than that of practically used LiPON glass is attractive. It is expected that a thin-film-type all-solid-state battery with Li_3BN_2 can be designed. However, when considering the application to a bulk-type all-solid-state battery, a conductivity of 10^{-5} S cm^{-1} is still low. The addition of other components is effective in improving the conductivity of glass electrolytes by using the mixed anion effect [34,35]. Optimization of the compositions of the nitride electrolytes with higher conductivity is now in progress.

In Fig. 8, the relative density, Young's modulus, and the conductivity of Li_3MX_a (M = B or P and X = S, N or O) are plotted against the electronegativity of element X in Li_3MX_a [36]. The Li_3BS_3 glass has

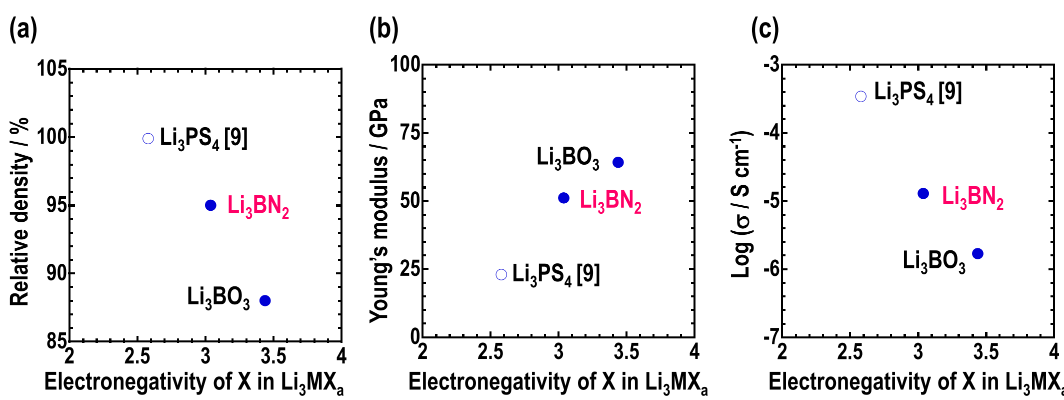

Fig. 8. Relationship between the electronegativity of X in Li_3MX_a and (a) the relative density, (b) Young's modulus, and (c) the conductivity of hot-pressed Li_3BN_2, Li_3BO_3, and Li_3PS_4.

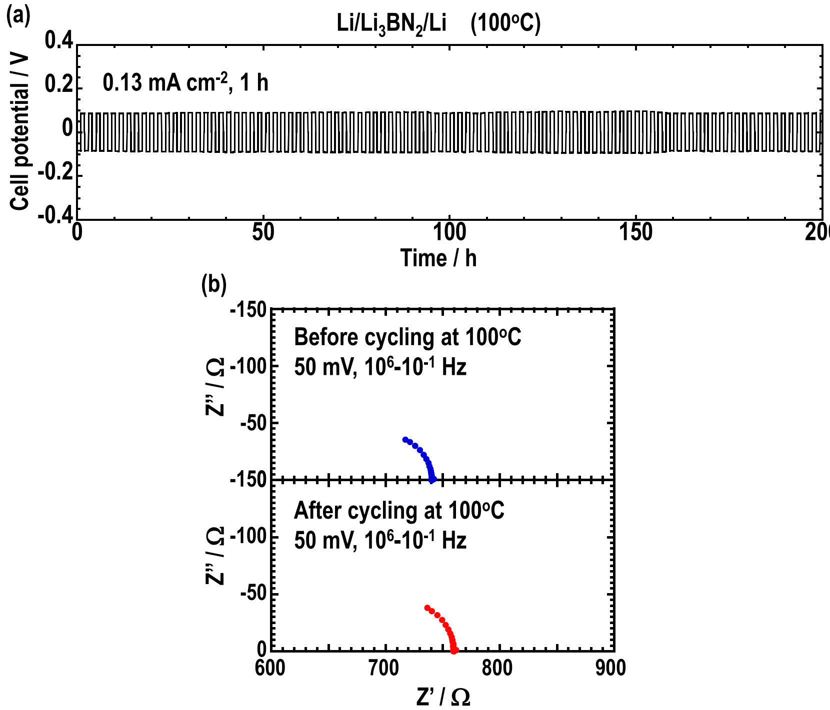

Fig. 9. (a) Galvanostatic cycling test of Li/Li_3BN_2 glass/Li cells at 100 °C. (b) Nyquist plots for the symmetric cells before and after the galvanostatic cycling test at 100 °C.

not yet been prepared by the mechanochemical synthesis. Thus, these properties of the Li_3PS_4 glass were plotted as a substitute for those of the Li_3BS_3 glass. It is considered that electronegativity of element X in Li_3MX_a (M = B or P and X = S, N or O) contributes to the relative density, Young's modulus, and the conductivity. Anions X are all in non-bridging states. The relative density of the hot-pressed Li_3MX_a glass electrolytes decreased with an increase in the electronegativity of element X. The lower the electronegativity of element X, the less the electrostatic interaction between Li^+ cation and MX_a^{3-} (PS_4^{3-}, BN_2^{3-}, and BO_3^{3-}) anion; in this situation, both Li^+ and MX_a^{3-} ions would diffuse readily, and the rearrangement of ions proceeds densification [10]. However, Young's modulus increased with the electronegativity of element X. The bond strength of M–X can be considered simply by the Coulomb interaction due to the difference in the electronegativity of element X. As the electronegativity of anion X increases, the bonding strength of M–X becomes high and Young's modulus increases. Although Young's modulus is not affected only by the electronegativity of element X, the electronegativity of element X is considered to have some influence on Young's modulus. In addition, the electronegativity of element X strongly affected the conduction property. It can be considered that the interaction between Li^+ and MX_a^{3-} decreased by decreasing the electronegativity of anion X, and thus Li^+ could be easily mobile. Further study is needed to confirm these tendencies.

In order to evaluate the electrochemical stability of Li_3BN_2 against a lithium metal negative electrode, long-term galvanostatic cycling tests in a symmetric cell (Li/Li_3BN_2/Li) were carried out at 100 °C. A constant current density of 0.13 mA cm^{-2} was applied for 1 h per half a cycle. As shown in Fig. 9a, the cell using the Li_3BN_2 glass electrolyte did not short-circuit after more than 100 cycles. The Nyquist plots for the symmetric cells before and after the galvanostatic cycling test at 100 °C are presented in Fig. 9b. No remarkable increase in resistance was observed even after the galvanostatic cycling test. These results indicate that the interface between lithium metal and Li_3BN_2 glass electrolyte has a good tolerance to the deposition-dissolution cycling of lithium at 100 °C. Detailed analyses for the interface between the Li_3BN_2 electrolyte and a typical oxide positive electrode are also important, and thus will be done in the near future.

4. Conclusions

A novel Li_3BN_2 glass electrolyte was developed. The conductivity of Li_3BN_2 glass is 1.3×10^{-5} S cm^{-1} at 25 °C, which is much higher than that of Li_3BO_3 oxide glass and a LiPON glass thin film. For the practical application of the nitride glass electrolyte, further increase in conductivity is needed. This is the first report on nitride glass electrolytes *via* the mechanochemical synthesis, and thus conductivity of the nitride-based glass electrolytes will be increased by optimizing elemental substitution. The nitride glass exhibited an excellent ductility like sulfide electrolytes. The Li symmetric cell using the Li_3BN_2 glass electrolyte was cycled stably at 100 °C without short-circuiting. We believe that this report will help to design a novel ionic conductor.

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